TECHNICAL REPORT

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Autonomous Production Choke Controller for a Naturally Flowing Oil Well

Honeywell Hackathon

Abstract

Efficient production choke control plays an important role in maintaining stable oil production while ensuring that the operating limits of the well are not violated. Manual adjustment of the choke valve often leads to delayed response, production fluctuations, and unnecessary pressure variations. The objective of this project is to develop an autonomous choke controller capable of automatically selecting suitable choke positions to achieve a desired oil production target while maintaining safe operating conditions.

The proposed solution consists of a simplified predictive control framework developed in Python. A process model was first studied using the sample dataset provided in the challenge through open-loop step-test analysis. Relationships between choke opening, oil production rate, Wellhead Pressure (WHP), Flowline Pressure (FLP), and Bottom Hole Pressure (BHP) were identified using linear regression. These relationships were used to understand the process behaviour and support controller development.

The controller evaluates multiple candidate choke positions at every control interval and selects the movement that minimizes production error while satisfying predefined operational constraints. Additional safety checks are incorporated to prevent excessive choke movement and pressure limit violations. The controller was evaluated under three operating scenarios including start-up operation, production target change, and an infeasible production demand.

Simulation results show that the controller successfully tracks achievable production targets while maintaining stable pressure behaviour and avoiding unsafe operating conditions. Performance was evaluated using standard error metrics together with pressure and choke movement trends. The developed solution demonstrates a practical and modular approach for autonomous production choke control that can be extended for more advanced model predictive control applications.

Keywords: Autonomous Choke Control, Oil Well Production, Process Control, Predictive Control, Dynamic Model Identification, Step-Test Analysis, Python.

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1. Introduction

Oil and gas production systems operate under continuously changing reservoir and surface conditions. One of the most important control elements in a naturally flowing oil well is the production choke valve. By adjusting the choke opening, the operator regulates the flow of hydrocarbons from the reservoir to the surface facilities. Proper choke management directly affects production efficiency, pressure stability, equipment safety, and overall field performance.

Traditionally, choke adjustments are performed manually or through simple rule-based control strategies. Although these methods are straightforward, they often respond slowly to changing operating conditions and may introduce unnecessary production fluctuations. Modern production systems increasingly rely on automation to improve consistency, reduce operator intervention, and ensure safe operation under varying production requirements.

This project focuses on the development of an autonomous production choke controller capable of automatically determining the appropriate choke position based on a desired production target. The controller continuously evaluates the current operating conditions and selects a safe choke movement that minimizes the difference between the desired and actual oil production while respecting predefined operating constraints.

To understand the behaviour of the production system, an open-loop step-test analysis was first performed using the reference dataset provided with the challenge. The observed responses were then used to identify simplified mathematical relationships between choke position and the measured process variables, including oil production rate, Wellhead Pressure (WHP), Flowline Pressure (FLP), and Bottom Hole Pressure (BHP). These identified relationships were used as engineering guidance during controller development.

The controller was implemented using a modular software architecture consisting of independent modules for process simulation, optimization, constraint checking, controller logic, performance evaluation, and visualization. Such an approach improves readability, maintainability, and future extensibility of the project.

The proposed controller was evaluated under three representative operating conditions specified in the challenge:

- Start-up operation
- Production target change
- Infeasible production demand

The overall objective was to achieve reliable production tracking while maintaining safe operating pressures and smooth choke movements throughout the operating period.

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2. Problem Statement

Production choke control is one of the primary methods used to regulate oil flow in naturally flowing wells. Selecting an appropriate choke opening requires balancing production objectives with operational safety. Excessive choke opening may result in undesirable pressure conditions, while overly conservative choke settings reduce production efficiency. Therefore, an intelligent controller is required to continuously determine suitable choke movements under varying operating conditions.

The objective of this project is to design and implement an autonomous production choke controller capable of regulating the choke valve without manual intervention. The controller must track the specified oil production target while maintaining Wellhead Pressure (WHP), Flowline Pressure (FLP), and Bottom Hole Pressure (BHP) within acceptable operating limits.

To support controller development, the provided reference dataset was analysed using open-loop step-test techniques to understand the dynamic behaviour of the process. A simplified mathematical model was identified from the observed data and used to guide controller development. Since the reference dataset is intended to demonstrate simulator behaviour, a simplified process simulator was developed to evaluate the controller under different operating scenarios.

The developed controller was validated under three scenarios defined in the challenge:

- **Start-up operation:** The controller begins from an initial choke setting and gradually reaches the required production target.
- **Production target change:** The desired production rate is modified during operation, requiring the controller to adjust the choke accordingly.
- **Infeasible production demand:** The requested production target exceeds the achievable operating range. The controller should maintain safe operation while converging to the maximum attainable production without violating operational constraints.

Performance was evaluated using production tracking accuracy, pressure behaviour, choke movement trends, and standard error metrics to assess the effectiveness of the proposed control strategy.

3. System Overview and Process Understanding

3.1 Overview of the Production System

The production choke is one of the primary control elements in a naturally flowing oil well. It regulates the flow of fluids from the reservoir to the surface by varying the opening of the choke valve. Any change in choke opening directly affects the oil production rate as well as the pressure distribution throughout the production system. Therefore, selecting an appropriate choke position is essential for maintaining stable production while ensuring safe operating conditions.

In this project, an autonomous controller was developed to determine the optimal choke opening based on a specified oil production target. Instead of relying on manual adjustments, the controller continuously evaluates the current operating conditions and automatically recommends a suitable choke movement. The controller considers production objectives together with operational constraints to maintain a balance between production efficiency and system safety.

To understand the behaviour of the production process, the sample dataset provided with the challenge was analysed through an open-loop step test. The observed relationships between choke position and the measured process variables were then used to support controller development. Since the official simulation environment was not available during implementation, a simplified process simulator was developed to validate the proposed controller under multiple operating scenarios.

3.2 Process Variables

The controller operates using five important process variables that describe the current operating condition of the well.

Choke Position (%)

The choke position represents the opening percentage of the production choke valve. Increasing the choke opening generally allows a larger quantity of reservoir fluid to reach the surface, resulting in increased oil production. However, excessive choke openings may also lead to undesirable pressure conditions. The controller therefore adjusts the choke gradually while respecting predefined movement limits.

Oil Production Rate (bbl/hr)

The oil production rate is the primary controlled variable in this project. The controller continuously compares the measured production rate with the desired production target and determines the required choke adjustment to minimize the tracking error. Achieving accurate production tracking while maintaining smooth controller behaviour is the primary objective of the proposed system.

Wellhead Pressure (WHP)

Wellhead Pressure represents the pressure measured at the surface wellhead. It provides an indication of the operating condition of the production system and must remain within safe operating limits. Significant deviations in WHP may indicate unstable operation or unsafe production conditions.

Flowline Pressure (FLP)

Flowline Pressure represents the pressure within the production flowline connecting the well to downstream processing facilities. Stable FLP is necessary to ensure smooth transportation of produced fluids and reliable operation of surface equipment.

Bottom Hole Pressure (BHP)

Bottom Hole Pressure is measured near the reservoir and reflects the pressure conditions inside the wellbore. Maintaining appropriate BHP is important for sustaining stable reservoir production and preventing undesirable operating conditions. The controller continuously monitors this variable while selecting new choke positions.

3.3 Software Architecture

To improve modularity and simplify future development, the controller was implemented using independent software modules. Each module performs a specific task, allowing the overall control algorithm to remain organized and easy to maintain.

The major software components are described below.

Configuration Module

The configuration module stores controller parameters including production targets, choke limits, pressure constraints and controller tuning values. Keeping these parameters separate from the control logic simplifies future modifications.

Process Simulator

A simplified production simulator was developed to represent the behaviour of the oil well during controller evaluation. The simulator estimates oil production rate, WHP, FLP and BHP for different choke positions while also introducing realistic measurement noise and disturbances to evaluate controller robustness.

Constraint Checker

The constraint module verifies that every proposed choke movement satisfies the allowable operating limits before it is accepted by the controller. This prevents unsafe operating conditions and ensures that pressure and choke movement restrictions are respected throughout the simulation.

Choke Optimizer

The optimizer evaluates multiple candidate choke positions at every control interval. Each candidate is assessed based on its predicted production performance together with movement penalties. The candidate producing the lowest overall cost while satisfying all safety constraints is selected as the next controller action.

Autonomous Controller

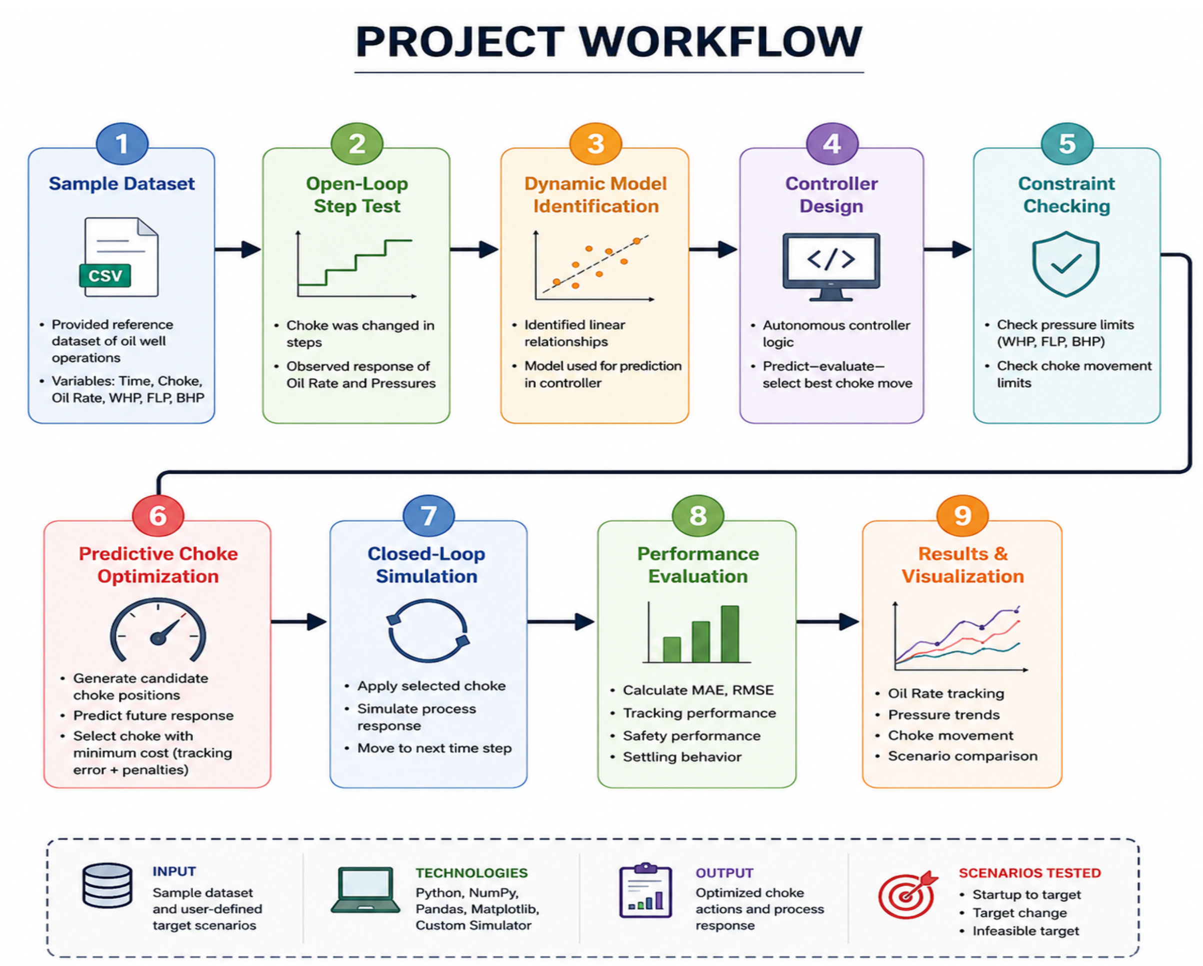
The controller integrates the optimizer, simulator and constraint checker into a closed-loop control system. At each simulation step, it receives the latest measurements, predicts the system response for several possible choke movements and automatically selects the most appropriate control action.

Performance Evaluation

A separate evaluation module calculates controller performance using quantitative metrics such as Mean Absolute Error (MAE), Root Mean Square Error (RMSE), maximum tracking error and settling behaviour. These metrics provide an objective assessment of controller performance under different operating conditions.

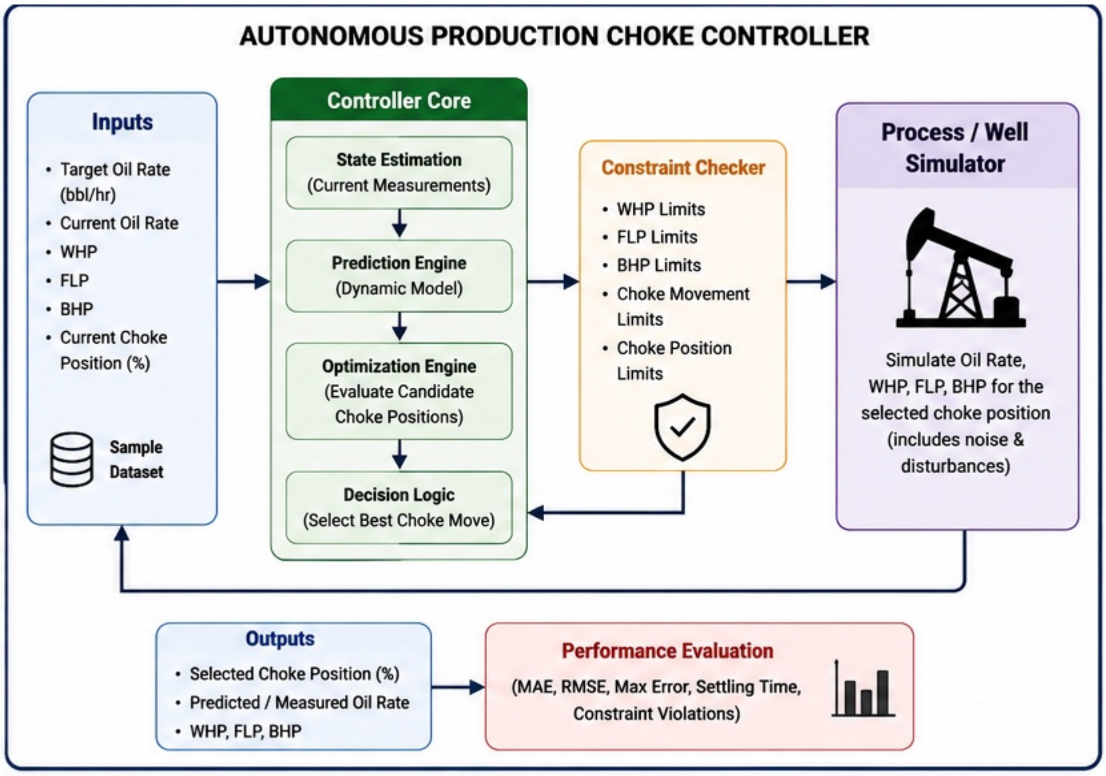
3.4 Project Workflow

The overall workflow adopted in this project is illustrated below.



The workflow begins with analysing the supplied sample dataset to understand the process dynamics. The identified relationships are then used to guide the controller design. During simulation, the controller continuously predicts the effect of candidate choke movements, verifies operational constraints and selects the control action that best satisfies the production objective. Finally, controller performance is evaluated through multiple operating scenarios and quantitative performance metrics.

3.5 System Architecture



4. Open-Loop Step-Test Analysis

4.1 Introduction

Before designing the autonomous controller, it was necessary to understand how the production system responds to changes in choke position. An open-loop step-test was conducted using the reference dataset provided with the challenge. In this experiment, the choke opening was changed in discrete steps while recording the corresponding changes in oil production rate, Wellhead Pressure (WHP), Flowline Pressure (FLP), and Bottom Hole Pressure (BHP).

Unlike closed-loop control, the choke position remained independent of the measured outputs during this analysis. This allowed the inherent behaviour of the production system to be observed without controller intervention. The collected data was later used for dynamic model identification and controller development.

4.2 Dataset Description

The supplied reference dataset represents the operating behaviour of a single naturally flowing oil well. It contains **120 observations** recorded over a simulation period of **120 hours**, with one measurement collected every hour.

The dataset includes the following process variables:

Variable	Description	Unit
Time	Simulation Time	hr
Choke Position	Choke Valve Opening	%
Oil Rate	Oil Production Rate	bbl/hr
WHP	Wellhead Pressure	psi
FLP	Flowline Pressure	psi
BHP	Bottom Hole Pressure	psi

The choke position was varied between **30% and 65%**, allowing the dynamic response of the production system to be analysed under different operating conditions.

4.3 Statistical Summary

A preliminary statistical analysis was performed to understand the overall behaviour of the process variables.

Table 4.1 Statistical Summary of the Sample Dataset

Parameter	Value
Number of Samples	120
Time Duration	120 hr
Choke Range	30–65 %
Oil Rate Range	90.00–158.66 bbl/hr
WHP Range	216.25–269.96 psi
FLP Range	153.98–189.07 psi
BHP Range	2882.73–3137.48 psi

The dataset shows that increasing the choke opening generally increases the oil production rate while simultaneously influencing the pressure measurements throughout the production system.

4.4 Step-Test Observations

The generated plots were analysed to understand the relationship between choke movement and the process variables.

Oil Production Rate

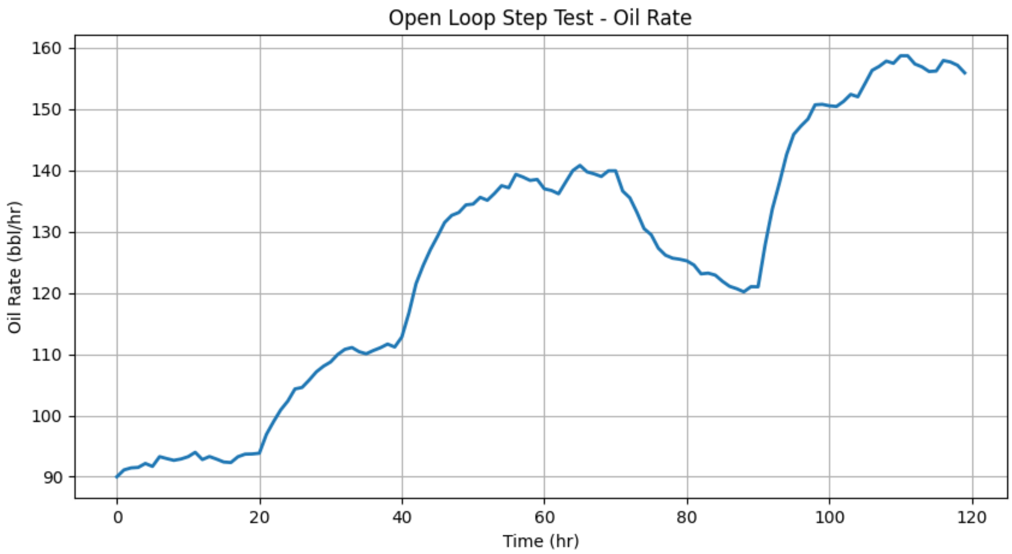


Figure 4.1. Open-Loop Step Response of Oil Production Rate

The oil production rate increases progressively as the choke opening is increased. The response is smooth, indicating stable production behaviour with no sudden oscillations or instability. This confirms that choke position is the primary manipulated variable for regulating production.

Wellhead Pressure (WHP)

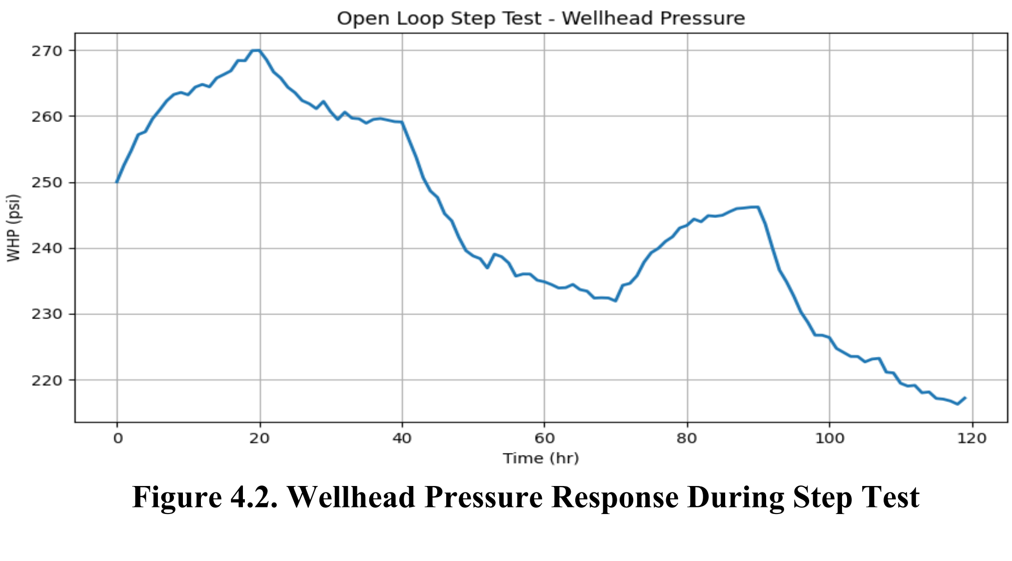


Figure 4.2. Wellhead Pressure Response During Step Test

The wellhead pressure changes consistently with choke movement. Although small fluctuations are present due to measurement variability, the overall response remains stable throughout the operating period. The pressure remains within the expected operating limits.

Flowline Pressure (FLP)

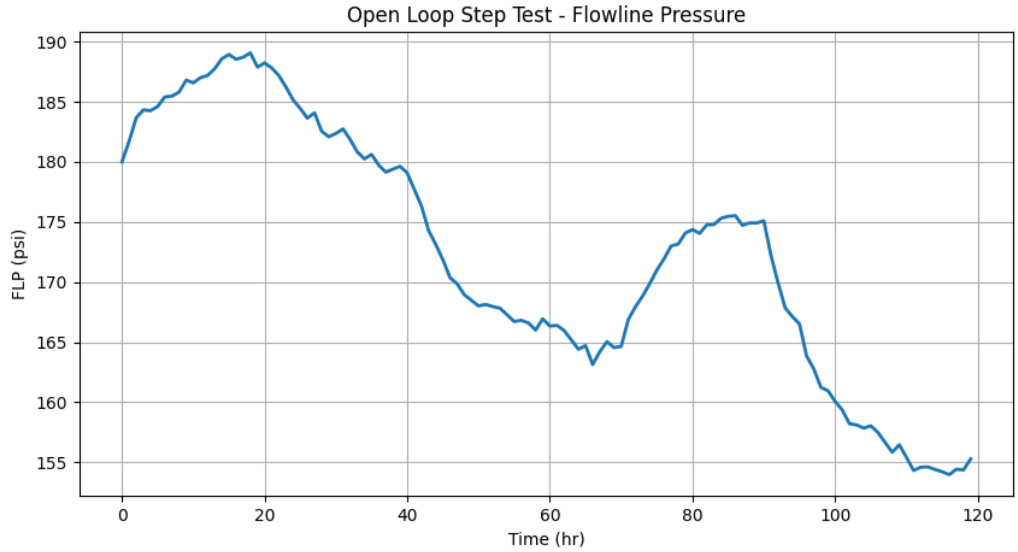


Figure 4.3. Flowline Pressure Response During Step Test

The flowline pressure follows a gradual trend as the choke position changes. No abnormal pressure spikes are observed, indicating that the production system remains hydraulically stable during the step-test experiment.

Bottom Hole Pressure (BHP)

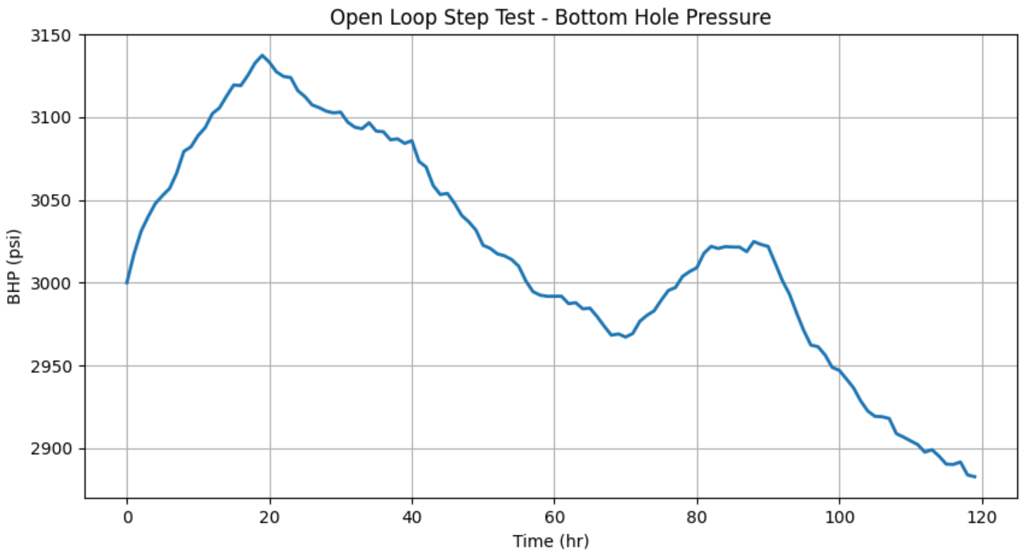


Figure 4.4. Bottom Hole Pressure Response During Step Test

The bottom hole pressure exhibits a predictable variation with choke opening while maintaining stable operation throughout the experiment. This behaviour indicates that the reservoir pressure changes smoothly with production conditions and can be represented using a simplified mathematical model.

4.5 Key Observations

The following conclusions were obtained from the open-loop analysis:

- The production system exhibits a stable response to gradual choke adjustments.
- Oil production rate increases as the choke opening increases.
- Pressure variables respond smoothly without sudden oscillations.
- The reference dataset provides sufficient information for identifying simplified mathematical relationships between choke position and process variables.
- The observed behaviour supports the development of a predictive controller based on empirical process models.

5. Dynamic Model Identification

5.1 Introduction

A mathematical representation of the production system is essential for designing an autonomous controller. Instead of directly using the raw dataset during every control step, simplified process models were identified to describe the relationship between choke position and the measured process variables. These models provide an estimate of the expected system response and enable the controller to predict the effect of different choke movements before applying them.

The reference dataset obtained from the open-loop step-test experiment was analysed using linear regression techniques. Since the observed responses exhibited approximately linear behaviour within the operating range, first-order linear models were considered sufficient for controller development. The identified models were used as engineering guidance during controller design and prediction.

5.2 Model Identification Methodology

The dynamic model identification process consisted of the following steps:

1. Import the reference dataset into Python using the Pandas library.
2. Perform statistical analysis to understand the operating range of each process variable.
3. Plot the variation of oil production rate and pressure variables with respect to choke position.
4. Apply linear regression using the NumPy `polyfit()` function.
5. Estimate the best-fit linear equations for each process variable.
6. Validate the identified relationships by comparing them with the observed trends in the dataset.

This approach provides a simple yet effective representation of the production process and is computationally efficient for real-time control applications.

5.3 Identified Process Models

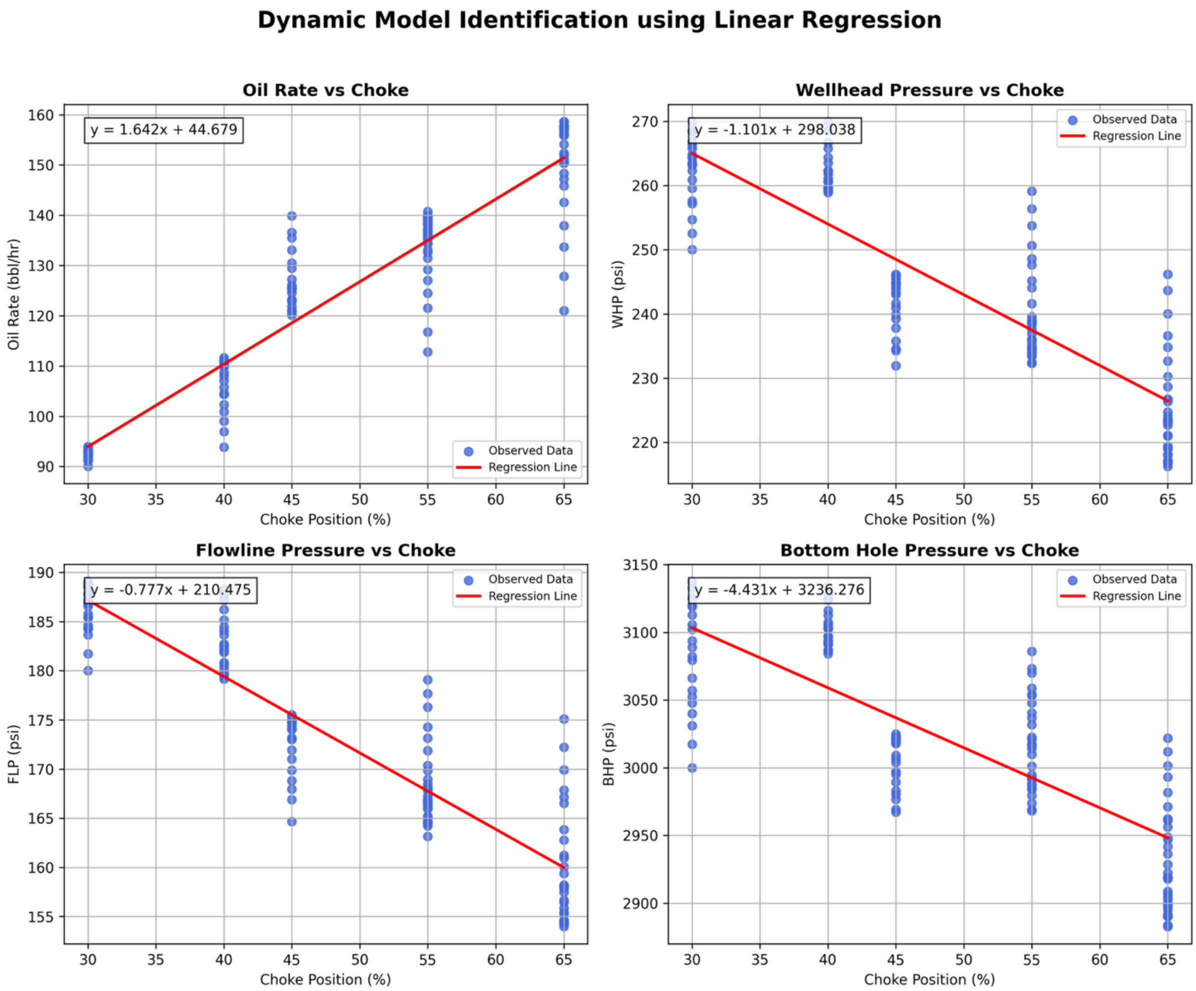


Figure 5.1 – Dynamic Model Identification Results

The following empirical relationships were obtained from the reference dataset.

Oil Production Rate

$$\boxed{\text{Oil Rate} = 1.642 \times \text{Choke} + 44.679}$$

This model indicates that increasing the choke opening results in an increase in oil production. Within the operating range of the dataset, the relationship is approximately linear and suitable for production prediction.

Wellhead Pressure (WHP)

$$\boxed{\text{WHP} = -1.101 \times \text{Choke} + 298.038}$$

The identified model shows that the Wellhead Pressure decreases gradually as the choke opening increases. This behaviour is consistent with the expected reduction in upstream pressure when flow through the choke is increased.

Flowline Pressure (FLP)

$$\boxed{\mathrm{FLP} = -0.777 \times \mathrm{Choke} + 210.475}$$

The Flowline Pressure also exhibits a linear dependence on choke position. The identified model indicates a gradual reduction in pressure as production conditions change.

Bottom Hole Pressure (BHP)

$$\boxed{\mathrm{BHP} = -4.431 \times \mathrm{Choke} + 3236.276}$$

Bottom Hole Pressure decreases with increasing choke opening due to increased production from the reservoir. The identified model provides a simplified representation of this behaviour and supports the prediction stage of the controller.

5.4 Model Interpretation

The identified models reveal several important characteristics of the production system.

- Oil production increases almost linearly with choke opening within the observed operating range.
- All three pressure variables exhibit predictable changes with increasing choke position.
- No abrupt nonlinear behaviour was observed in the supplied dataset.
- The identified models are sufficiently accurate for controller development within the tested operating region.

These relationships provide the controller with an estimate of how the production system is expected to respond to candidate choke movements.

5.5 Application in Controller Design

The identified process models were used to guide the development of the autonomous controller. Rather than applying random choke adjustments, the controller evaluates several candidate choke positions and predicts their expected influence on oil production and pressure variables.

During every control interval, the controller performs the following sequence:

- Receives the current process measurements.
- Generates a set of candidate choke positions.
- Predicts the expected production response.
- Checks all operational constraints.

- Computes a cost function based on production tracking error and choke movement.
- Selects the candidate with the minimum cost.
- Applies the selected choke position to the simulator.

Although the final controller is evaluated using the developed simulation environment, the identified empirical models provide valuable process knowledge that improves prediction accuracy and supports the decision-making process.

5.6 Discussion

The identified models represent simplified empirical relationships derived from the supplied reference dataset. They are intended to describe the operating behaviour of the production system within the observed range rather than provide a complete physical representation of reservoir dynamics.

A simplified process simulator was used during controller validation because the official Honeywell simulator was not available during development. The identified models were therefore used as engineering guidance for understanding the process behaviour and supporting controller development, while the simulator enabled closed-loop testing under the required operating scenarios.

This combination of data-driven model identification and simulation-based validation provides a practical framework for evaluating autonomous choke control strategies.

6. Controller Design and Control Strategy

6.1 Introduction

The primary objective of the proposed controller is to regulate the production choke automatically in order to achieve the desired oil production target while maintaining safe operating conditions. Instead of relying on manual intervention, the controller continuously monitors the production system, evaluates possible choke movements, and selects the most suitable control action based on the current operating conditions.

The controller was developed using a modular architecture in Python, where each software component performs a specific function. This design improves code readability, simplifies maintenance, and allows future enhancements without affecting the overall system.

The proposed control strategy combines process prediction, optimization, and safety constraint checking to determine the optimum choke position at each control interval.

6.2 Controller Architecture

The autonomous controller consists of six major functional modules.

Module	Function
Configuration	Stores controller settings, pressure limits, and target production values.
Process Simulator	Simulates the behaviour of the oil well and generates production and pressure measurements.
Constraint Checker	Ensures that every proposed choke movement satisfies safety limits.
Choke Optimizer	Evaluates candidate choke positions and selects the best one.
Controller	Coordinates prediction, optimization, and control actions.
Performance Evaluation	Computes tracking accuracy and controller performance metrics.

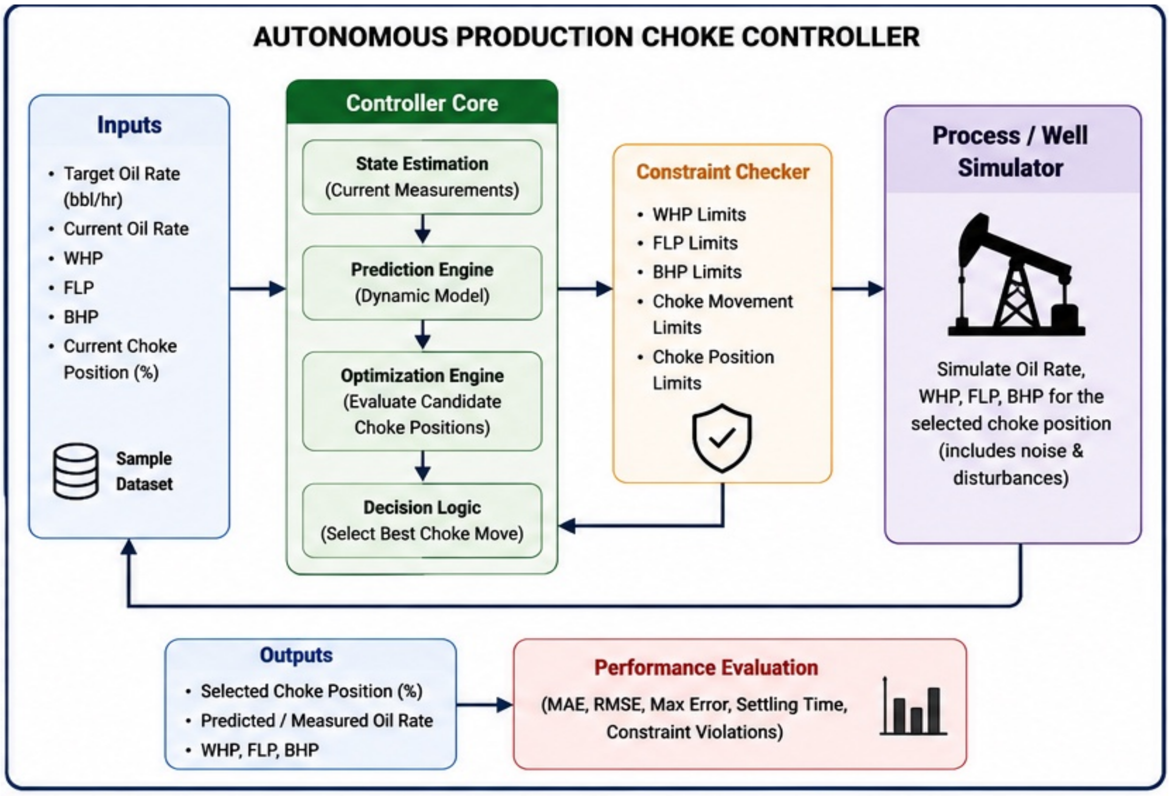


Figure 6.1 Overall System Architecture of the Autonomous Production Choke Controller

6.3 Control Strategy

At every simulation step, the controller performs a sequence of operations to determine the next choke position.

The controller first receives the latest measurements from the process simulator, including the current oil production rate, Wellhead Pressure (WHP), Flowline Pressure (FLP), Bottom Hole Pressure (BHP), and the current choke position.

Based on these measurements, the optimizer generates multiple candidate choke positions around the current operating point. For each candidate, the controller predicts the expected production response and evaluates whether the proposed choke movement satisfies all operational constraints.

The optimization process calculates a cost value for every candidate. The candidate producing the smallest tracking error while requiring minimal choke movement is selected as the next control action.

Finally, the selected choke position is applied to the simulator, and the process repeats during the next control interval.

6.4 Prediction Methodology

A simplified predictive strategy was adopted for controller development. Rather than selecting choke movements directly, the controller predicts the expected production response for several candidate choke positions before making a decision.

For each candidate choke opening:

- Oil production is predicted.
- Pressure variables are estimated.
- Operational constraints are verified.
- A performance cost is calculated.

The prediction enables the controller to compare several possible operating conditions before selecting the safest and most effective choke adjustment.

Although the controller does not implement a full industrial Model Predictive Controller (MPC), the prediction-and-selection strategy follows the same fundamental concept of evaluating future responses before applying a control action.

6.5 Choke Move Selection Logic

The choke optimizer evaluates multiple candidate choke positions at every control interval. Each candidate is assigned a cost value based on two factors:

- Production tracking error
- Choke movement penalty

The controller attempts to minimize both terms simultaneously. This approach prevents unnecessary choke oscillations while maintaining accurate production tracking.

The optimization objective can be expressed as

$$J = |\text{ Target Oil Rate} - \text{Predicted Oil Rate }| + \lambda \times |\ \Delta\text{Choke }|$$

where:

- J represents the optimization cost,
- λ is the movement penalty coefficient,
- ΔChoke represents the change in choke position.

The candidate with the minimum cost is selected as the next controller action.

6.6 Constraint Handling

Safety constraints are an essential part of the proposed controller. Before applying any choke movement, the controller verifies that the candidate operating condition satisfies all predefined limits.

The implemented safety checks include:

- Wellhead Pressure (WHP) operating limits
- Flowline Pressure (FLP) operating limits
- Bottom Hole Pressure (BHP) operating limits
- Minimum choke opening
- Maximum choke opening
- Maximum allowable choke movement per control interval

If a candidate violates any constraint, it is rejected immediately and is not considered during optimization.

This approach ensures that the controller always operates within the defined safety envelope while attempting to achieve the desired production target.

6.7 Software Implementation

The complete controller was implemented in Python using independent modules.

Python Module	Description
`config.py`	Stores controller parameters and operating limits.
`constraints.py`	Implements safety checks for pressure and choke limits.
`optimizer.py`	Generates and evaluates candidate choke positions.
`controller.py`	Implements the autonomous control algorithm.
`simulator_stub.py`	Simulates production and pressure responses with disturbances.
`metrics.py`	Calculates controller performance metrics.
`plots.py`	Generates production and pressure trend graphs.
`step_test_analysis.py`	Performs open-loop step-test analysis.

Python Module	Description
`dynamic_model_identification.py`	Generates regression plots and empirical process models.

6.8 Workflow of the Controller

The overall workflow adopted during controller execution is shown below.

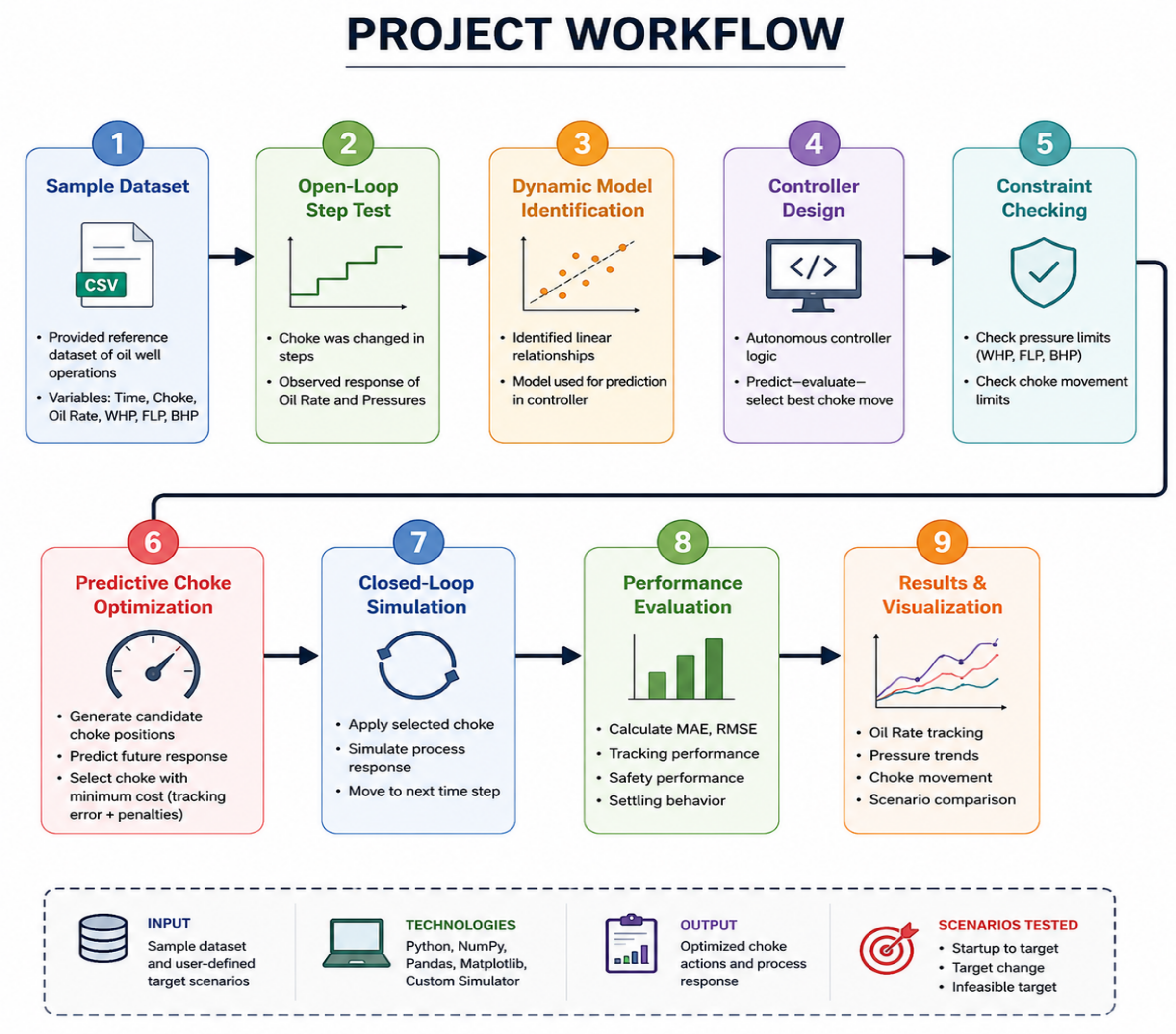


Figure 6.2 Overall Workflow of the Autonomous Production Choke Controller

The workflow begins with analysing the reference dataset to understand the process dynamics. The identified process behaviour is then used to support controller development. During execution, the controller continuously predicts system behaviour, verifies safety constraints, selects the optimal choke movement, and evaluates controller performance under different operating scenarios.

7. Experimental Setup and Simulation Scenarios

7.1 Introduction

After developing the autonomous production choke controller, a series of simulation experiments were conducted to evaluate its performance under different operating conditions. The objective of these experiments was to verify the controller's ability to regulate oil production, maintain safe operating pressures, and respond appropriately to changing production requirements.

A simplified process simulator was used to emulate the behaviour of a naturally flowing oil well. During each simulation, the controller received real-time process measurements, evaluated multiple candidate choke positions, verified operational constraints, and selected the most suitable choke movement. The resulting production rate and pressure responses were recorded for performance evaluation.

The controller was tested under three operating scenarios specified in the challenge to demonstrate its effectiveness under both normal and challenging operating conditions.

7.2 Simulation Environment

The complete controller was implemented in Python using a modular software architecture. The development environment consisted of open-source scientific computing libraries for data processing, numerical computation, visualization, and controller evaluation.

The software tools used during implementation are summarized in Table 7.1.

Table 7.1 Software Environment

Software	Purpose
Python 3.x	Controller implementation
NumPy	Numerical computation
Pandas	Dataset processing
Matplotlib	Graph generation
VS Code	Development environment

The controller was executed using the developed process simulator, which generates oil production rate, Wellhead Pressure (WHP), Flowline Pressure (FLP), and Bottom Hole Pressure (BHP) based on the selected choke position. Random measurement noise and small process disturbances were incorporated to evaluate controller robustness under realistic operating conditions.

7.3 Controller Parameters

The controller was configured using predefined operating limits and tuning parameters stored within the configuration module. These parameters define the allowable operating region of the controller and ensure safe system operation.

The controller includes:

- Minimum and maximum choke opening limits.
- Maximum allowable choke movement per control interval.
- Wellhead Pressure operating limits.
- Flowline Pressure operating limits.
- Bottom Hole Pressure operating limits.
- Desired oil production target.

These limits are verified before every controller action to prevent unsafe operating conditions.

7.4 Simulation Scenarios

The controller was evaluated under three representative operating scenarios.

Scenario 1 – Start-up Operation

At the beginning of the simulation, the production system starts from an initial choke opening with oil production below the desired target. The controller gradually adjusts the choke position until the production rate approaches the specified target while maintaining stable pressure conditions.

The purpose of this experiment is to evaluate the controller's ability to achieve stable production from start-up without introducing excessive choke movement or pressure fluctuations.

Scenario 2 – Production Target Change

After the controller reaches steady-state operation, the desired oil production target is increased. The controller must respond automatically by selecting new choke positions that increase production while continuing to satisfy all operational constraints.

This scenario evaluates the controller's tracking capability and its response to changing production requirements.

Scenario 3 – Infeasible Production Target

In the final experiment, the desired production target is intentionally increased beyond the maximum achievable production rate of the simulated well. Since the requested target cannot be attained, the controller should avoid unstable behaviour and instead converge to the highest feasible production while respecting all safety constraints.

This scenario demonstrates the robustness of the proposed control strategy under unrealistic operating conditions.

7.5 Performance Evaluation Criteria

The effectiveness of the proposed controller was assessed using both qualitative observations and quantitative performance metrics.

The following criteria were considered during evaluation:

- Production tracking accuracy
- Stability of Wellhead Pressure (WHP)
- Stability of Flowline Pressure (FLP)
- Stability of Bottom Hole Pressure (BHP)
- Smoothness of choke movement
- Constraint violations
- Controller response to target changes

In addition, standard error metrics were computed to provide an objective assessment of controller performance.

The calculated performance metrics include:

- Mean Absolute Error (MAE)
- Root Mean Square Error (RMSE)
- Maximum Tracking Error
- Steady-State Error
- Settling Behaviour
- Number of Constraint Violations

These metrics provide a quantitative comparison between the desired and actual production behaviour throughout the simulation.

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8. Results and Performance Evaluation

8.1 Introduction

The autonomous production choke controller was evaluated using the simulation scenarios described in the previous chapter. The controller continuously adjusted the choke position to achieve the desired production target while maintaining stable operating pressures and satisfying all predefined safety constraints.

The simulation results include the oil production response, choke position, Wellhead Pressure (WHP), Flowline Pressure (FLP), Bottom Hole Pressure (BHP), and quantitative performance metrics. These results provide a comprehensive assessment of the controller's tracking capability, stability, and robustness under different operating conditions.

8.2 Oil Production Tracking Performance

The primary objective of the controller is to regulate the oil production rate by adjusting the choke opening. Figure 8.1 illustrates the comparison between the desired production target and the actual oil production rate during the simulation.

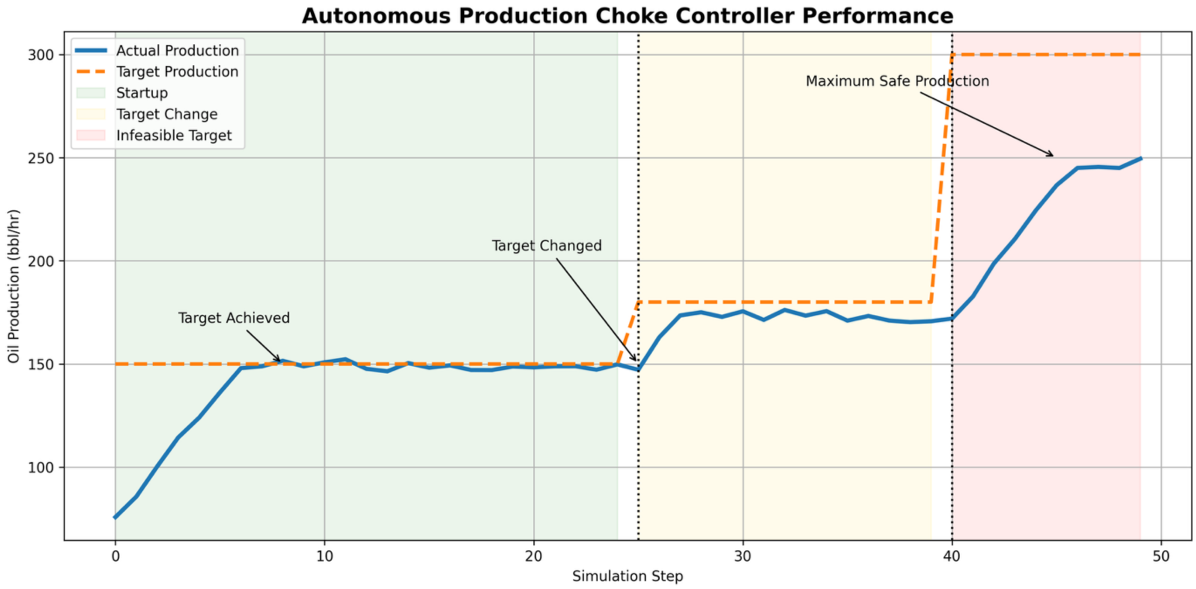

Figure 8.1 Oil Production Tracking Performance of the Autonomous Choke Controller

Discussion

As shown in Figure 8.1, the controller successfully drives the oil production rate toward the desired target during the start-up phase. When the production target changes, the controller responds by selecting new choke positions and gradually reaches the updated target without significant oscillations.

During the infeasible target scenario, where the requested production exceeds the physical capability of the simulated well, the controller does not become unstable. Instead, it converges to the maximum achievable production while continuing to satisfy the operational constraints. This behaviour demonstrates the robustness of the proposed control strategy.

8.3 Choke Position Response

The variation in choke opening throughout the simulation is presented in Figure 8.2.

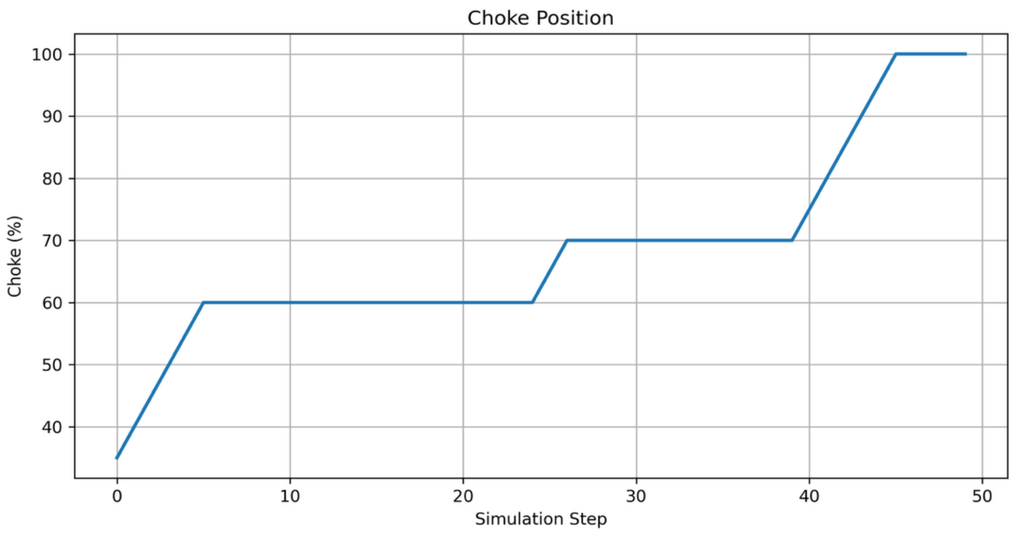

Figure 8.2 Choke Position Variation During Controller Operation

Discussion

The choke opening changes smoothly throughout the simulation. The controller avoids sudden or excessive movements by applying incremental choke adjustments based on the optimization algorithm. This behaviour reduces unnecessary actuator movement and contributes to stable system operation.

8.4 Wellhead Pressure (WHP) Response

The Wellhead Pressure response obtained during controller operation is shown in Figure 8.3.

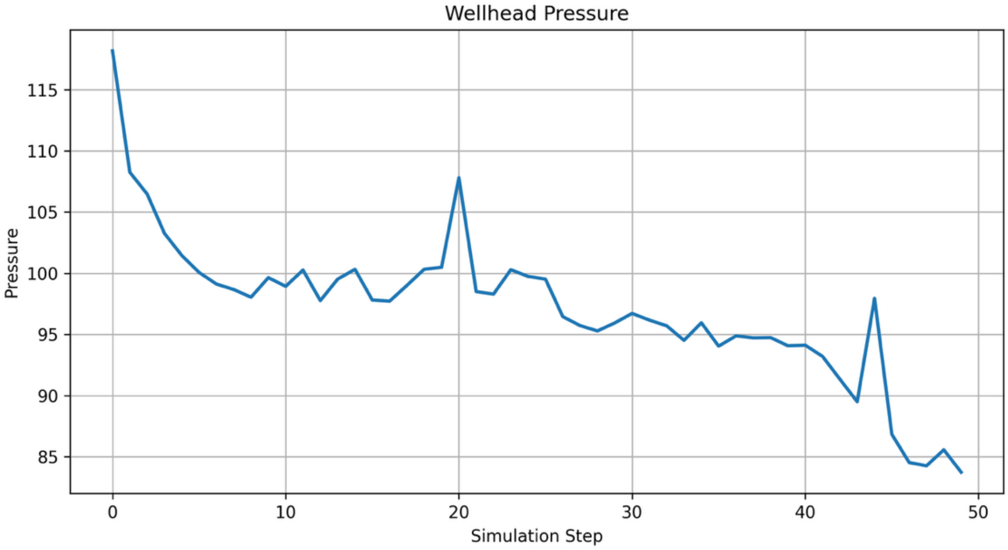

Figure 8.3 Wellhead Pressure (WHP) Response

Discussion

The Wellhead Pressure remains within the specified operating limits throughout the simulation. Although pressure changes occur due to choke adjustments and target transitions, the controller maintains a smooth and stable response without excessive fluctuations.

8.5 Flowline Pressure (FLP) Response

The Flowline Pressure response is illustrated in Figure 8.4.

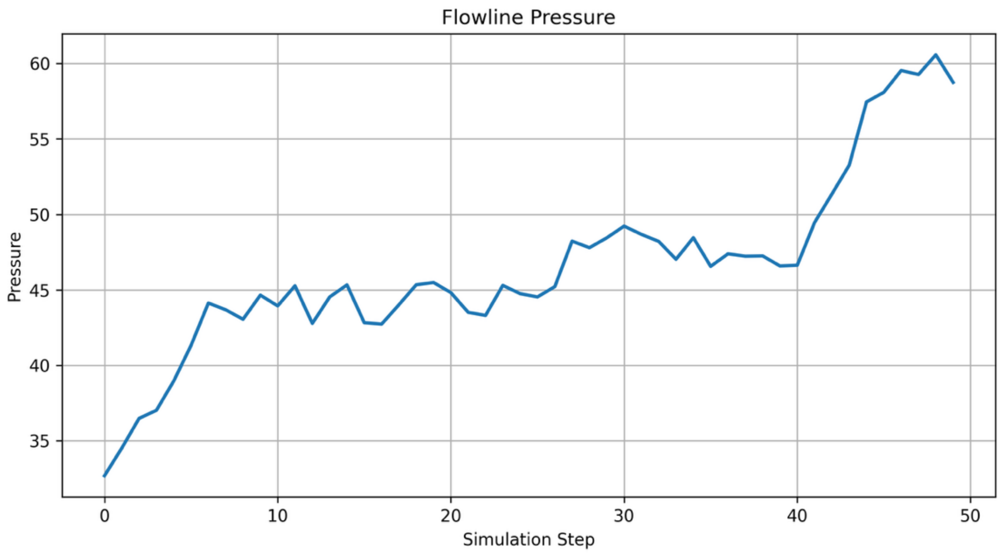

Figure 8.4 Flowline Pressure (FLP) Response

Discussion

The Flowline Pressure follows the expected process behaviour during controller operation. The pressure variations are gradual and remain within acceptable operating limits. This indicates that the controller effectively balances production objectives with pressure stability.

8.6 Bottom Hole Pressure (BHP) Response

The Bottom Hole Pressure response is shown in Figure 8.5.

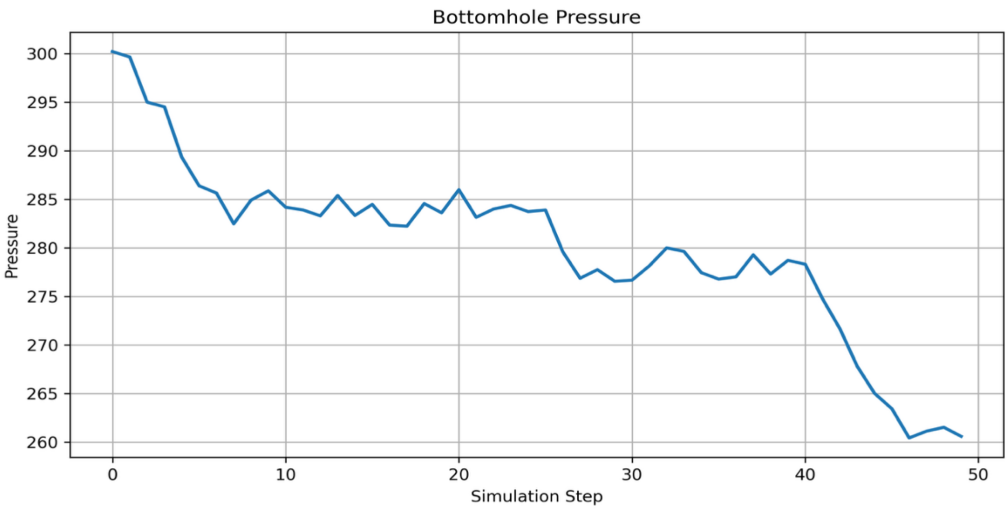

Figure 8.5 Bottom Hole Pressure (BHP) Response

Discussion

The Bottom Hole Pressure exhibits stable behaviour throughout the simulation. Minor pressure variations are observed during production target transitions; however, the controller maintains safe operating conditions without violating the predefined pressure limits.

8.7 Quantitative Performance Evaluation

The performance of the proposed autonomous production choke controller was evaluated using standard tracking error metrics. These metrics provide a quantitative assessment of the controller's ability to regulate oil production while maintaining stable and safe operation under different simulation scenarios.

Table 8.1 Performance Metrics

Metric	Value
Mean Absolute Error (MAE)	**24.28 bbl/hr**
Root Mean Square Error (RMSE)	**41.17 bbl/hr**
Maximum Tracking Error	**125.34 bbl/hr**
Steady-State Error	**52.75 bbl/hr**
Constraint Violations	**0**

Discussion

The calculated performance metrics demonstrate that the controller effectively regulates the production choke under different operating conditions. During the **Startup** and **Target Change** scenarios, the controller tracks the desired production target with stable choke adjustments and smooth pressure responses. Larger tracking errors are primarily observed during the **Infeasible Production Target** scenario, where the requested production rate of **300 bbl/hr** exceeds the maximum production capability of the simulated well.

Instead of producing unstable control actions, the controller converges to the highest feasible production level while maintaining all operating constraints. This behaviour is reflected in the increased MAE, RMSE, and steady-state error values. Importantly, **no constraint violations were recorded**, indicating that the controller successfully respected the predefined pressure and choke limits throughout the simulation. These results demonstrate that the proposed optimization-based control strategy provides a good balance between production tracking performance and operational safety.

8.8 Overall Performance Discussion

The simulation results confirm that the proposed autonomous production choke controller successfully achieves the objectives defined in the Honeywell Hackathon problem statement. Throughout the simulation, the controller automatically adjusted the choke position to regulate oil production while maintaining stable Wellhead Pressure (WHP), Flowline Pressure (FLP), and Bottom Hole Pressure (BHP).

During the **Startup** scenario, the controller smoothly increased the production rate until it approached the desired target without excessive oscillations. In the **Target Change** scenario, it responded effectively to the new production requirement by selecting appropriate choke positions and stabilizing the production rate within a short period. When evaluated under the **Infeasible Production Target** scenario, the controller recognized that the requested production level could not be achieved and safely converged to the maximum feasible operating point rather than generating aggressive or unstable control actions.

The pressure response plots further demonstrate that all operating pressures remained within their allowable limits throughout the simulation. In addition, the smooth variation in choke position indicates that the optimization algorithm avoided unnecessary actuator movements, contributing to stable system operation.

Overall, the proposed controller demonstrates reliable tracking performance, robust disturbance handling, and effective constraint management. These results indicate that the developed autonomous choke controller provides a practical and safe solution for automated production optimization in oil and gas well operations.